A weak-star-convergent-basis obstruction for a polyhedral Lindenstrauss problem

Candidate partial result for arXiv:2303.10023

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Status. Candidate partial result, likely valid, requiring human review. Problem 6.8 remains open in general. The theorem below rules out every ℓ_1-predual for which the standard dual basis is weak-star convergent.

1 Source problem

De Bernardi proves that every polyhedral Lindenstrauss space admits arbitrarily close equivalent polyhedral norms whose unit balls are generated by their extreme points. He then asks [1, Problem 6.8]:

> Does there exist a polyhedral Lindenstrauss space E satisfying
>
> $$B_E = \overline{\mathrm{conv}}(\mathrm{ext}(B_E))?$$

As throughout the source paper, "space" is understood to be infinite-dimensional unless otherwise stated.

For $f = (f_n) \in \ell_1$, put

$$W_f = \left\{ x = (x_n) \in c : \lim_{n\to\infty} x_n = \sum_{n=1}^{\infty} f_n x_n \right\}, \tag{1}$$

with the supremum norm inherited from c. Hyperplanes of this form are a standard source of explicit ℓ_1-preduals; the polyhedral example in [2] is obtained with $f_n = 2^{-n}$.

2 Proof intuition

Suppose the standard basis (e_n^*) of $X^* = \ell_1$ converges weak-star to f. Evaluating $x \in X$ on these basis vectors embeds X isometrically into c. The weak-star limit imposes precisely the single equation defining W_f, and an annihilator calculation shows that the range is all of W_f.

The extreme-point calculation is then coordinatewise. If two coordinates of $x \in B_{W_f}$ are strictly inside $[-1, 1]$, a small two-coordinate perturbation can be chosen to preserve the defining equation. Thus an extreme point has at most one unsaturated coordinate. Its limit must consequently be $s \in \{-1, 1\}$. The equality

$$1 = |s| = \left|\sum f_n x_n\right| \le \sum |f_n||x_n| \le \|f\|_1 \le 1$$

6.2. **Lindenstrauss spaces.** A Banach space E is called an L_1*-predual space* or a *Lindenstrauss space* if its dual is isometric to $L_1(\mu)$ for some measure μ. The following result about polyhedral Lindenstrauss spaces holds.

Theorem 6.6 ([3, Theorem 4.3])**.** *Let E be a Lindenstrauss space. The following properties are equivalent:*

(i) *E is a polyhedral space;*
(ii) *E does not contain an isometric copy of c;*
(iii) *E is* (V)*-polyhedral.*

The theorem above implies that if E is a polyhedral Lindenstrauss space then condition $(**)$ is satisfied, hence we have the following corollary.

Corollary 6.7. *Let E be a polyhedral Lindenstrauss space. Then, for each $\varepsilon > 0$, E admits an ε-equivalent polyhedral renorming such that the corresponding closed unit ball is the closed convex hull of its extreme points.*

In view of the previous result, the following problem arises.

Problem 6.8. Does there exist a polyhedral Lindenstrauss space E satisfying $B_E = \overline{\operatorname{conv}}\,(\operatorname{ext}(B_E))$?

Figure 1: Problem 6.8 and its immediate context in [1, p. 20], rendered and cropped from the local source PDF.

forces $\|f\|_1 = 1$ and $x_n = s\,\operatorname{sgn}(f_n)$ on the support of f; in fact every coordinate is saturated. If the support of f has two points, all convex combinations of extreme points obey a fixed coordinate equation that an elementary vector of B_{W_f} violates. If the support has only one point, W_f is just an isometric copy of the nonpolyhedral space c.

3 The hyperplane representation

Lemma 1. *Let X be a real Banach space and fix an isometric identification $X^* = \ell_1$. If the standard basis (e_n^*) converges in $\sigma(\ell_1, X)$ to $f \in \ell_1$, then $\|f\|_1 \le 1$ and X is linearly isometric to W_f.*

Proof. Weak-star lower semicontinuity of the norm gives $\|f\|_1 \le 1$. Define

$$J : X \longrightarrow c, \qquad Jx = (e_n^*(x))_{n=1}^{\infty}.$$

The sequence is convergent because $e_n^* \to f$ weak-star. Moreover, $\operatorname{ext}(B_{X^*}) = \{\pm e_n^* : n \in \mathbb{N}\}$ is a boundary of X, so

$$\|Jx\|_\infty = \sup_n |e_n^*(x)| = \|x\|.$$

Thus J is an isometry with closed range $Y = JX$. For $x \in X$,

$$\lim_n (Jx)_n = f(x) = \sum_{n=1}^{\infty} f_n e_n^*(x) = \sum_{n=1}^{\infty} f_n (Jx)_n,$$

so $Y \subseteq W_f$.

Use the standard representation

$$c^* = \mathbb{R} \oplus_1 \ell_1, \qquad (a_0, a)(z) = a_0 \lim_n z_n + \sum_n a_n z_n.$$

Such a functional annihilates Y exactly when

$$a_0 f + a = 0 \quad \text{in } X^* = \ell_1.$$

Consequently

$$Y^\perp = \operatorname{span}\{(1, -f)\}.$$

But $W_f = \ker(1, -f)$ has the same annihilator. Two closed subspaces with the same annihilator coincide, hence $Y = W_f$. $\square$

4 The extreme-point calculation

Write $\operatorname{supp} f = \{n : f_n \neq 0\}$.

Lemma 2. *Let $f \in \ell_1$ with $\|f\|_1 \le 1$. If $\|f\|_1 < 1$, then $\operatorname{ext}(B_{W_f}) = \varnothing$. If $\|f\|_1 = 1$, the extreme points are exactly the sequences $x \in c$ for which there is $s \in \{-1, 1\}$ such that*

$$x_n = s \operatorname{sgn}(f_n) \qquad (n \in \operatorname{supp} f), \tag{2}$$

$$x_n \in \{-1, 1\} \qquad (n \in \mathbb{N}), \tag{3}$$

$$x_n = s \qquad \textit{eventually.} \tag{4}$$

In particular, this set is empty unless $f_n \ge 0$ for all sufficiently large $n \in \operatorname{supp} f$.

Proof. Fix $x \in B_{W_f}$ and let

$$U = \{n : |x_n| < 1\}.$$

If U contains two distinct indices i, j, then x is not extreme. Indeed, if one of f_i, f_j is zero, a sufficiently small nonzero multiple of the corresponding coordinate vector gives $h \in c_0$ with $\sum_n f_n h_n = 0$ and $x \pm h \in B_c$. If both coefficients are nonzero, use instead a sufficiently small multiple of

$$f_j e_i - f_i e_j.$$

Again $h \in c_0$, $\sum_n f_n h_n = 0$, and $x \pm h \in B_c$. In either case $h \in W_f$, so x is a nontrivial midpoint in B_{W_f}. The same one-coordinate perturbation shows that if $U = \{j\}$ for an extreme point, then necessarily $f_j \neq 0$.

It follows that an extreme x has at most one unsaturated coordinate. Since x converges, all but at most one of its coordinates lie in $\{-1, 1\}$, and therefore $\lim_n x_n = s$ for some $s \in \{-1, 1\}$ and $x_n = s$ eventually. The defining equation for W_f now gives

$$1 = |s| = \left| \sum_n f_n x_n \right| \le \sum_n |f_n||x_n| \le \|f\|_1 \le 1.$$

Every inequality is an equality. Hence $\|f\|_1 = 1$, every coordinate with $f_n \neq 0$ is saturated, and all nonzero summands $f_n x_n$ have sign s. Thus (2) holds. The only possible unsaturated coordinate would now have $f_j = 0$, contradicting the necessary condition above; hence (3) also holds. This proves the necessity of (2)–(4), and also proves that there are no extreme points when $\|f\|_1 < 1$.

Conversely, a sequence satisfying these three conditions lies in W_f: the equality case gives $\sum_n f_n x_n = s \sum_n |f_n| = s = \lim_n x_n$. It is an extreme point of the larger cube B_c, because every coordinate is 1 or -1. It is therefore extreme in the subset B_{W_f} as well. $\square$

5 The obstruction

Theorem 3. *Let X be an infinite-dimensional real ℓ_1-predual for which, under some isometric identification $X^* = \ell_1$, the standard basis is weak-star convergent. Then X cannot satisfy both*

1. *X is polyhedral, and*
2. *$B_X = \overline{\mathrm{conv}}(\mathrm{ext}(B_X))$.*

Consequently no such X answers Problem 6.8 of [1].

Proof. By Lemma 1, X is isometric to W_f for some $f \in B_{\ell_1}$. If $\|f\|_1 < 1$, Lemma 2 gives $\mathrm{ext}(B_{W_f}) = \varnothing$, so condition (2) fails.

Suppose $\|f\|_1 = 1$ and $\mathrm{supp}\, f$ contains distinct indices i, j. Every extreme point x satisfies

$$\mathrm{sgn}(f_i)x_i = \mathrm{sgn}(f_j)x_j \tag{5}$$

by Lemma 2. Hence the closed convex hull of all extreme points is contained in the closed hyperplane defined by (5). On the other hand, a sufficiently small nonzero multiple y of

$$f_j e_i - f_i e_j$$

belongs to B_{W_f}, since it has limit zero and $\sum_n f_n y_n = 0$. Its two signed coordinates in (5) have opposite signs, so it does not belong to that hyperplane. Thus condition (2) again fails.

It remains to consider $\mathrm{supp}\, f = \{k\}$. Then $f = \sigma e_k$ for $\sigma \in \{-1, 1\}$ and

$$W_f = \{x \in c : \lim_n x_n = \sigma x_k\}.$$

Deleting the kth coordinate identifies W_f isometrically with c; the inverse reinserts $\sigma \lim_n x_n$ in that coordinate. The space c is not polyhedral. For a direct verification, let

$$u_n = \cos(1/n), \qquad v_n = \sin(1/n).$$

Both u, v belong to c. On $F = \mathrm{span}\{u, v\}$ the coordinate functionals correspond to $p_n = (\cos(1/n), \sin(1/n)) \in \mathbb{R}^2$, together with their limit $p_\infty = (1, 0)$. Each p_n is exposed in

$$\overline{\mathrm{conv}}\{\pm p_n : n \in \mathbb{N}\}$$

by the Euclidean functional p_n, because its scalar product with every other $\pm p_m$ and with $\pm p_\infty$ is strictly less than 1. This set is the polar of B_F, so B_F cannot be a polygon. Hence c, and therefore W_f, is not polyhedral. Condition (1) fails in the last case. □

Corollary 4. *No hyperplane $W_f \subset c$ with $\|f\|_1 \le 1$ is an infinite-dimensional polyhedral Lindenstrauss space whose given unit ball is the closed convex hull of its extreme points. In particular, the explicit polyhedral hyperplane with $f_n = 2^{-n}$ from [2] has only the two extreme points $\pm(1, 1, \dots)$ and cannot answer Problem 6.8.*

Proof. The first assertion is the case of Theorem 3 represented by Lemma 1; alternatively it follows directly from Lemmas 2 and the proof of the theorem. For $f_n = 2^{-n}$, the support is all of $\mathbb{N}$ and every sign is positive, so Lemma 2 leaves exactly the two constant sign sequences. □

6 Verification, scope, and search status

The argument is noncomputational. Its load-bearing points were checked separately in the accompanying `verification.md`: the boundary norm identity in Lemma 1, the annihilator calculation, all zero-coefficient cases in the perturbation argument, equality in the ℓ_1 triangle inequality, and the one-coordinate-support reduction to c.

This is a partial result only. It does not cover an ℓ_1-predual for which the dual atoms cannot be enumerated as one weak-star-convergent sequence, nonseparable L_1-preduals, or more general simplex spaces. It does not answer the source's renorming Problem 6.3 or cardinality Problem 6.10.

Bounded literature check. On 11 August 2026, the run indexes and the local full-source corpus were searched by arXiv id and the terms "polyhedral Lindenstrauss", "hyperplane of c", "weak-star convergent basis", and "closed convex hull of its extreme points". External exact-title, exact-question, DOI, and close-keyword searches found the source paper and the earlier hyperplane papers [2, 3], but no answer to Problem 6.8 and no statement of Theorem 3. OpenAlex listed no citing work for the 2024 journal article at the time of the check. This is a bounded novelty audit, not a claim of definitive priority.

Human-review recommendation. Review Lemma 1's use of the extreme dual boundary and the equality $Y^\perp = \operatorname{span}(1, -f)$, then check that the perturbations in Lemma 2 cover every possible unsaturated coordinate pattern. If those steps pass, the obstruction theorem is ready to be treated as a rigorous partial result.

References

[1] C. A. De Bernardi, *Unit balls of polyhedral Banach spaces with many extreme points*, Studia Math. **275** (2024), 175–196; arXiv:2303.10023, doi:10.4064/sm230710-31-12.

[2] E. Casini, E. Miglierina, and Ł. Piasecki, *A remark on spaces of affine continuous functions on a simplex*, arXiv:1503.09088 (2015).

[3] E. Casini, E. Miglierina, Ł. Piasecki, and L. Veselý, *Rethinking polyhedrality for Lindenstrauss spaces*, Israel J. Math. **216** (2016), 355–369; arXiv:1506.08559, doi:10.1007/s11856-016-1412-8.